MATLAB | karthikkumarp2416sse-bit/5G-... | MIMO System BER Analysis | (28) WhatsApp | Ask Gemini

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Title | Subtitle | X-Label | Y-Label | Z-Label | Legend | Colorbar | Grid | X-Grid | Y-Grid | LABELS AND ANNOTATIONS

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Files

Name

- 5G AND 6G LAB
- LAB EXPERIMENT
- New Folder
- ASSEMENT QNO 1.m
- EX : 2.m
- ex 1.m
- EX 2 ..m
- EX 2..m
- Ex 2.m
- ex 5.m
- ex1.m

Workspace

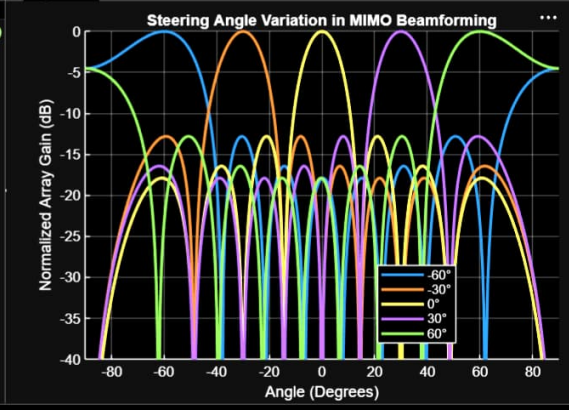
| Name | Value | Size | Class |
|---|---|---|---|
| AF | 1×361 double | 1×361 | double |
| k | 361 | 1×1 | double |
| m | 5 | 1×1 | double |
| N | 8 | 1×1 | double |
| phase | [0;0.4209;0.8... | 8×1 | double |
| steeringA... | [-60,-30,0,30,... | 1×5 | double |
| theta | 1×361 double | 1×361 | double |
| theta0 | 60 | 1×1 | double |

2.m | ex3.m | EX4.m | untitled4.m | EX6.m | EX7.m | EX8.m

/MATLAB Drive/EX8.m

```
17      for k = 1:length(theta)
18
19          phase = pi*(0:N-1)'*(sind(theta(k)) - sind(theta0));
20          AF(k) = abs(sum(exp(1j*phase)));
21
22      end
23
24      % Normalize
25      AF = AF/max(AF);
26
27      % Plot in dB
28      plot(theta,20*log10(AF),'LineWidth',2);
29
30  end
31
32  grid on;
33  xlabel('Angle (Degrees)');
34  ylabel('Normalized Array Gain (dB)');
35  title('Steering Angle Variation in MIMO Beamforming');
36  legend('-60°','-30°','0°','30°','60°','Location','best');
37  axis([-90 90 -40 0]);
```

Figure 1




Command Window

>>

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